

EXPERIMENTAL_SETUP_v2

Glassy Carbon Electrode Integration into Honeycomb-Serpentine Moiré System
Dual-Variable θ Control: Optical (twist angle) + Electrical (DC bias)

Honeycomb Serpentine Optics Series · Chur · 2026 | Follows EXPERIMENTAL_SETUP_v1 (N20 motor / polarising film series)

Overview

This document describes the next experimental phase of the Honeycomb Serpentine Moiré optics series. The core upgrade is glassy carbon coating applied to the honeycomb structure embedded within the serpentine scaffold. The carbon layer serves simultaneously as (1) an optical mask that modulates Moiré fringe contrast via selective absorption/reflection, and (2) a DC electrode connected via a battery holder after resistance verification with a multimeter.

This dual role enables independent control of two components of the effective axion-like phase field: θ_{optical} set by the input polarising film twist angle, and θ_{electric} set by the applied DC voltage. The combined system constitutes a tuneable table-top analogue of axion electrodynamics.

1. Experimental Architecture

The optical stack is unchanged from v1 except for the addition of the carbon-coated honeycomb electrode layer between the input polarising film and the rotating output film.

Layer	Component	Role	Variable
① Input	Fixed polarising film	Sets base θ_{optical}	Rotation angle ϕ_{in}
② Active	Honeycomb-Serpentine + glassy carbon coating	Optical mask + DC electrode	Voltage V (battery)
③ Output	Rotating polarising film (N20 motor)	Generates dynamic Moiré fringe	Motor speed ω
④ Detector	Camera	Records fringe dynamics	Frame rate, exposure

The key insight is that the carbon layer couples optical and electrical degrees of freedom at the same spatial location — the honeycomb nodes — so both θ components act on identical fringe geometry. This is essential for a clean dual-variable experiment.

2. Glassy Carbon Coating — Properties and Rationale

2.1 Why Glassy Carbon?

Optical properties

Electrical properties

Glassy carbon has broadband absorption in the visible range (reflectance ~5–10%, absorptance ~60–70% at 500 nm). Coating the honeycomb ribs selectively darkens the lattice boundaries, increasing Moiré fringe contrast without introducing birefringence or polarisation artefacts.

Why not metal?

Metal coatings (Al, Au) introduce specular reflection that creates secondary interference patterns unrelated to the Moiré geometry. Glassy carbon scatters diffusely — it absorbs rather than redirects light.

Resistivity: 40–60 $\mu\Omega\cdot\text{m}$ (semi-metallic). A continuous coating on a honeycomb rib network of ~30–100 μm feature size will typically yield sheet resistance $R_{\text{sq}} = 10\text{--}500\ \Omega/\blacksquare$ depending on coating thickness. This is easily measured with a two-probe multimeter across the scaffold.

Biocompatibility note

Glassy carbon is chemically inert and biocompatible — relevant if the experiment later extends to cell-modulation applications as outlined in the immunology research framework.

2.2 Expected Resistance Values

Before connecting the battery holder, verify coating continuity with a multimeter in resistance mode. Expected ranges:

Measurement point	Expected R	Interpretation
Across single honeycomb cell (~1 mm)	100 Ω – 5 k Ω	Continuous coating, usable electrode
Across full serpentine (~5 cm)	1 k Ω – 50 k Ω	Network resistance, normal
Open circuit (OL / ∞)	—	Coating discontinuous — recoat
< 10 Ω across full scaffold	—	Possible short — check for bridging

3. Step-by-Step Experimental Protocol

Phase A — Resistance Verification (Go/No-Go)

Prerequisites

Glassy carbon coated honeycomb-serpentine scaffold (dried/cured)

Digital multimeter with two-probe leads

Two conductive contact points prepared at opposite ends of the scaffold

- Step 1. Set multimeter to resistance (Ω) mode, 20 k Ω range.

Step 2. Touch probes to the two prepared contact points on the carbon coating.

Step 3. Record resistance R_0 . If OL $\rightarrow$ coating is discontinuous, do not proceed to Phase B.

Step 4. Map resistance at 3–5 points across the scaffold to check uniformity (variation < 5 $\times$ is acceptable).

Step 5. Photograph multimeter display + probe positions for GitHub documentation.
- If $R_0 < 10\ \Omega$, inspect for carbon bridges across intended gaps before proceeding.

Phase B — Optical Mask Characterisation (No Voltage)

- Step 1. Insert carbon-coated scaffold into the optical stack between input film and output film.

Step 2. Run motor at standard speed (same as v1 baseline).

Step 3. Record 60-second video: compare fringe contrast with v1 baseline (no carbon).

- Step 4.** Extract brightness time series using OpenCV pipeline (same as v1).
- Step 5.** Measure: peak-to-trough contrast ratio, fringe spatial frequency, cluster detachment frequency f_{detach} .
- Step 6.** Expected result: higher contrast fringes, f_{detach} unchanged (carbon is passive at $V=0$).

Phase C — DC Electrode Activation (Battery Holder)

Equipment

AA or AAA battery holder (1×1.5 V to 3×1.5 V = 4.5 V max for initial tests)

Two thin wire leads soldered or clipped to the carbon contact points

Multimeter in parallel to monitor actual voltage across scaffold during recording

- Step 1.** Connect battery holder leads to carbon contact points. Verify polarity is consistent across all runs.
- Step 2.** Measure voltage V_{actual} across scaffold with multimeter (will be < battery EMF due to R_0).
- Step 3.** Record 60-second video for each voltage level: 0 V (disconnected), 1.5 V, 3.0 V, 4.5 V.
- Step 4.** Between recordings, allow 30 seconds for thermal equilibration.
- Step 5.** For each voltage, extract: fringe contrast, f_{detach} , FFT spectrum (0–2 Hz range).
- Step 6.** Document: battery voltage, V_{actual} , R_0 , motor speed, input film angle ϕ_{in} .

■ Keep $V_{\text{actual}} < 5\text{ V}$ for initial runs. Higher voltages risk delamination of thin carbon coatings and may introduce Joule heating artefacts in the fringe pattern.

Phase D — Dual-Variable θ Sweep

Once Phase C confirms a voltage-dependent fringe response, systematically vary both control variables to map the θ surface:

Run	ϕ_{in} (input film angle)	V_{actual} (DC bias)	Observable target
D-01	0° (parallel)	0 V	Baseline — maximum transmission
D-02	0°	1.5 V	Electric θ only
D-03	22.5°	0 V	Optical θ only
D-04	22.5°	1.5 V	Combined θ — additive?
D-05	45°	0 V	Optical $\theta = \pi/4$
D-06	45°	1.5 V	Combined θ
D-07	90° (crossed)	0 V	Maximum extinction — $\theta = \pi/2$
D-08	90°	1.5 V	Does electric field break extinction?

4. Mapping to Axion-like θ Framework

The dual-variable control maps cleanly onto the effective axion phase field:

$$\theta_{\text{eff}}(r, V) = \theta_{\text{optical}}(r, \phi_{\text{in}}) + \theta_{\text{electric}}(V)$$
$$\theta_{\text{optical}} = 2 \cdot \phi_{\text{in}} \text{ [from Malus's law: } T = \cos^2(\Delta\phi)\text{]}$$
$$\theta_{\text{electric}} = \alpha \cdot V / V_{\text{ref}} \text{ [linear approximation, } \alpha \text{ to be fitted]}$$

θ_{optical} — what it controls

The input film angle ϕ_{in} sets the base transmission envelope of the Moiré fringe. At $\phi_{\text{in}} = 0^\circ$ the lattice is maximally transparent ($\theta = 0$); at $\phi_{\text{in}} = 90^\circ$ it is maximally opaque ($\theta = \pi$). This is the geometric, time-independent component of the axion field — analogous to the crystal structure of a topological insulator.

θ_{electric} — what it controls

The DC voltage modifies the local permittivity of the carbon-coated honeycomb network, shifting the effective optical path length and thus the fringe phase. This is the dynamic, externally tuneable component — analogous to applying an external magnetic field in an axion haloscope (e.g. ADMX). Its coefficient α is the primary quantity to extract from Phase D.

Key prediction: satellite-to-cluster transition boundary

In v1 (no carbon, no voltage), the satellite-to-cluster transition was triggered by removing the input-side film ($\theta_{\text{optical}} \rightarrow$ unconstrained). In v2, re-inserting the input film pins θ_{optical} . The prediction is that applying $V \geq V_{\text{crit}}$ can re-trigger the transition purely electrically — i.e., θ_{electric} alone can cross the domain-wall nucleation threshold. If confirmed, this is a direct electrical analogue of the axion field driving a topological phase transition.

Observable signature: f_{detach} jumps discontinuously at $V = V_{\text{crit}}$ (look for hysteresis when ramping V up vs down).

5. Data Recording and Analysis Pipeline

5.1 Video and Metadata

- Filename convention: v2_PhaseX_phiYY_VZZ.mp4 (e.g. v2_PhaseD_phi45_V15.mp4 for $\phi=45^\circ$, $V=1.5$ V)
- Compress with ffmpeg: H.264 CRF 28 / preset fast / AAC 96k (same as v1)
- Metadata JSON per run: {phase, phi_in_deg, V_battery, V_actual, R_0, motor_pwm, frame_rate, timestamp}

5.2 Analysis (reuse v1 pipeline)

- OpenCV brightness/contrast time series extraction (per-frame ROI mean)
- FFT + STFT in 0–2 Hz range; track f_{detach} peak position vs V and ϕ_{in}
- Peak detection for cluster formation/detachment events
- New in v2: plot $f_{\text{detach}}(V)$ at fixed ϕ_{in} to extract θ_{electric} coefficient α
- New in v2: 2D surface plot of contrast ratio as function of (ϕ_{in}, V)

5.3 GitHub Repository Structure (v2 additions)

```
honeycomb-serpentine-moire/
■■■ EXPERIMENTAL_SETUP_v2.md ← this document (markdown version)
■■■ data/
■ ■■■ v1/ (existing)
■ ■■■ v2/
■ ■■■ phase_A/ resistance verification photos + values
■ ■■■ phase_B/ optical mask baseline videos
■ ■■■ phase_C/ voltage sweep videos
■ ■■■ phase_D/ dual-variable sweep videos
■■■ analysis/
■ ■■■ v1_pipeline.py (existing)
■ ■■■ v2_dual_theta.py NEW: dual-variable  $\theta$  analysis
■■■ README.md update: add v2 section
```

6. Additional Bill of Materials (v2 only)

Item	Specification	Purpose	Est. cost
Glassy carbon coating	Pyrolysed SU-8 or CVD amorphous C, 30–100 μm feature size	Optical mask + electrode	Lab process
Battery holder	AA $\times$ 1–3 (1.5–4.5 V), with leads	DC bias source	< \$2
Digital multimeter	Any 2-probe, Ω range to 50 k Ω	Resistance verification	Existing
Thin wire leads	28–30 AWG, insulated	Connect holder to scaffold	< \$1
Conductive paint/paste	Silver paste or carbon paste, for contact pads	Reliable probe contact points	< \$5

7. Summary and Next Steps

The v2 upgrade converts the Honeycomb Serpentine Moiré system from a purely optical instrument into a dual-variable θ laboratory. Glassy carbon coating on the honeycomb structure simultaneously enhances fringe contrast (optical mask) and provides a DC electrode for electrical phase control. The four-phase protocol — resistance check, optical baseline, voltage sweep, dual-variable sweep — is designed to be reproducible and fully documentable on GitHub.

Immediate next actions

1. Apply glassy carbon coating to honeycomb-serpentine scaffold.
2. Phase A: multimeter resistance check → record R_0 and uniformity map.
3. Phase B: re-insert input-side polarising film → confirm v1 fringe recovery.
4. Phase C: connect battery holder → record fringe response at 0 / 1.5 / 3.0 / 4.5 V.
5. Commit all videos, metadata JSON, and resistance photos to GitHub under data/v2/.
6. Run v2_dual_theta.py analysis → extract α coefficient and V_{crit} estimate.

Document history

v1 — N20 motor polarising-film series: cluster formation/detachment quasi-periodic dynamics ($f \approx 0.108$ Hz), satellite-to-cluster transition observed upon removal of input-side privacy film.

v2 (this document) — glassy carbon electrode integration, dual-variable θ_{optical} + θ_{electric} control, Phase A–D protocol.